

Week 2 — Claims Data Field Guide

Medical and pharmacy claim anatomy • code systems • Rx panel vocabulary • data-quality checklist • de-identification rules

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Student reference — keep it next to every claims analysis this semester

Companion to [Pharma_Commercial_Datasets_Reference.docx](#) (the nine-category taxonomy) and the [Week 1 glossary](#)

How to use this guide. Part 1 tells you what each field on a claim means and which stakeholder it encodes. Part 2 is the vocabulary of Rx panel data and syndicated reports. Part 3 is the checklist you run before trusting any extract. Part 4 is the cohort-definition template used in class and required in your term-project data plan. Part 5 is what the law and the license allow you to do with patient-level data. Vendor names are examples, not endorsements.

Part 1 • Anatomy of a claim

1.1 The medical claim (professional 837P, institutional 837I)

A medical claim is what a provider (physician, clinic, hospital) sends to a payer to be paid for care. One claim has one header and one or more service lines. Vendor extracts usually flatten the header onto every line, so header fields repeat.

Field (typical extract name)	Level	What it is	Analyst's use	Gotchas
claim_id	Header	Payer/clearinghouse claim number	De-duplication; resubmissions share a root ID	Suffixes/versions for corrections; keep the latest
patient_id (token)	Header	De-identified patient key	Longitudinal linkage	Tokens differ across vendors unless a common tokenization is licensed
service_date (from/to)	Header + line	Date(s) of service	Index dates, sequencing, LOT	Institutional claims span days; use line dates for drugs
billing_provider_npi / tax_id	Header	Organization paid	HCO / IDN attribution, 340B eligibility	Type 2 NPI ≠ the clinician
rendering_provider_npi	Line	Clinician who performed the service	Prescriber/treater attribution, targeting	Missing when billed under a group; locums
place_of_service (POS)	Header	Site of care code	Office (11) vs hospital outpatient (22/19) vs inpatient (21)	Payment and 340B differ by POS; telehealth codes 02/10
diagnosis_code_1...12 (ICD-10-CM)	Header	Why the patient was seen; dx1 = primary	Cohorts, comorbidity, staging proxies	Rule-out coding; dx copied to drug lines; annual code changes
procedure_code (CPT / HCPCS)	Line	What was done or given	Drug administrations (J/Q-codes), infusions (96413), E/M visits (9921x), imaging, biopsies	New drugs sit in unclassified codes (J3490, C9399) for months
units	Line	Quantity of the procedure/drug unit	Dose reconstruction (J9312 = 10 mg per unit)	Unit definitions differ per code; JW modifier = wastage
modifiers	Line	Two-character qualifiers	25 (separate E/M), 59 (distinct service), JW (waste), JZ (no waste)	Vendor may drop them
revenue_code	Line (facility)	Hospital department/category	0636 drugs requiring detail coding; 0450 ER; 0250 pharmacy	Only on institutional claims
DRG	Header	Bundled payment group	Inpatient episodes	Drugs inside a DRG have

	(inpatient)	for a stay		no line — invisible
billed / allowed / paid / patient_resp	Header + line	Money at each step	Cost of care; site-of-care economics	Says nothing about manufacturer net price; negative lines = adjustments
payer / plan / channel	Header	Who paid	Commercial vs Medicare vs Medicaid cuts	Channel is a vendor crosswalk; Medicare Advantage often coded as Commercial

1.2 The pharmacy claim (NCPDP D.0)

A pharmacy claim is the real-time adjudication between the pharmacy and the PBM at the moment of dispensing. One row per event: a paid claim, a rejection, or a reversal (a paid claim later cancelled — usually because the patient never picked it up).

Field	What it is	Analyst's use	Gotchas
rx_id / claim_reference	Adjudication event ID	De-duplication; pairing reversals to the paid claim	A paid + reversed pair nets to zero
patient token • birth year • sex • ZIP3	De-identified patient	Linkage, demographics, geography	Ages ≥ 90 collapsed; ZIP3 may be suppressed
fill_date	Date dispensed (adjudicated)	Persistence, gaps, refills	Mail order fills 90 days; normalize by days_supply
NDC (11-digit)	Product × labeler × package	Drug, strength, brand/generic, market definition	10-digit source formats; NDC recycling; biosimilars separate
days_supply • quantity	Coverage of the fill	Adherence (PDC), 30-vs-90 normalization	Quantity units vary (tablets, mL, pens)
prescriber_npi	Who wrote it	Targeting, specialty, share of prescriber	Group NPIs; NPs/PAs under supervising MD
pharmacy_id • type	Where dispensed	Channel (retail, mail, specialty, LTC)	PBM-owned pharmacies steer volume
plan / BIN-PCN-group • payer_type	Benefit that paid	Access analysis; channel mix	Cash fills and copay-card-only fills may be absent
claim_status • reject_code	PAID / REJECTED / REVERSED; NCPDP reject code	Prior auth (75), not covered (70), refill too soon (79), step therapy	Multiple rejections per attempt; reversals within days
ingredient_cost • dispensing_fee • patient_pay • plan_pay • copay_assistance	Money at the counter	Out-of-pocket analysis, abandonment	Not the manufacturer's net; rebates are elsewhere
DAW code • refill_number • new/refill flag	Substitution and refill status	NRx vs TRx	NRx ≠ new-to-brand (needs longitudinal token)

1.3 The code systems

Code system	Format	Identifies	Maintained by	Notes
ICD-10-CM	Letter + 2 digits + up to 4 characters (C83.30, E11.65)	Diagnoses	CDC/NCHS with CMS; updated every Oct 1	Specificity varies; first three characters = category; procedure codes (ICD-10-PCS) are separate and inpatient-only
CPT	5 digits (99214, 96413)	Physician services and procedures	AMA	E/M levels reflect documentation, not severity; Category II/III codes exist
HCPCS Level II	Letter + 4 digits (J9312, Q2041, G0463)	Drugs, biologics, supplies, some services	CMS	J-codes = drugs; Q-codes = temporary (CAR-T, biosimilars); C-codes = hospital outpatient; new products start unclassified
NDC	10 digits (4-4-2, 5-3-2, 5-4-1) → normalized 11 (5-4-2)	Drug product, labeler, package	FDA (labeler assigns segments)	Same molecule = many NDCs; map NDC → molecule → market with a drug master
NPI	10 digits, Luhn check, prefix 80840	Type 1 individuals; Type 2 organizations	CMS NPPES (free download)	Taxonomy codes give specialty; enrich with commercial masters
POS	2 digits	Place of service	CMS	11 office • 19 off-campus

				outpatient · 21 inpatient · 22 on-campus outpatient · 23 ER · 02/10 telehealth
MS-DRG	3 digits	Inpatient stay group	CMS	Drugs bundled; use for episode cost, not drug utilization
LOINC / SNOMED / RxNorm	Numeric / alphanumeric	Lab tests / clinical concepts / normalized drug names	Regenstrief / SNOMED International / NLM	EHR and lab data, not claims; RxNorm bridges NDC → ingredient
DEA number	2 letters + 7 digits	Prescriber registration for controlled substances	DEA	Sometimes used as an alternate HCP key in older Rx feeds; NPI is the standard join
NCPDP reject codes	2–3 characters	Why a pharmacy claim was denied	NCPDP	70 not covered · 75 prior auth · 76 plan limit · 79 refill too soon · 88 DUR reject

Part 2 · Rx panel data and syndicated reports — the vocabulary

Term	Meaning	Why it matters
Panel	The set of pharmacies (or clearinghouses, payers) that supply data to a vendor	Everything you see is a sample; the panel changes over time
Coverage	Share of a stratum's dispensing volume the panel captures	Uneven: chains > independents; mail 2–3 facilities; specialty closed
Projection	Scaling panel counts to the universe with stratum factors (universe volume ÷ panel volume)	Stratified ratio estimation; uncovered cells are imputed
Projection factor	1 / coverage for a stratum	Large factors = fragile numbers; single-member strata are the risk
Universe	The census of outlets or prescribers being projected to	Built from wholesaler data, NPPES, state boards
Restatement	Revision of prior periods as late claims arrive or the panel changes	Never compare a fresh period with a mature one
Cross-sectional	Counts by outlet/prescriber × product × period, no patient key	Share, trend, targeting; cannot do persistence or switching
Longitudinal (patient-level, LRx / APLD)	Events linked over time by a de-identified patient token	Journeys, NBRx, persistence, cohorts, LOT
Patient capture	Share of a patient's care that the data sees	Biases rates (persistence, time-to-event); report it next to every rate
Open claims	Claims observed in transit (clearinghouse/switch); no enrollment	Breadth and speed; unknown gaps
Closed claims	A payer's complete adjudication with enrollment/eligibility	Denominators you can trust; narrower and slower
TRx / NRx / NBRx	Total scripts / new scripts (no fill of this product at this pharmacy in ~12 months) / new-to-brand patients	NRx is cross-sectional; NBRx needs a token
Source of business	Where new-to-brand patients came from (naïve, switch, add-on)	Needs longitudinal data
DDD / 867 / 852	Sales-out (867) and inventory (852) EDI feeds from wholesalers	Outlet × NDC × week; reconciles to revenue; not patient demand
Xponent-style / NPA-style	Projected prescriber-level Rx / national audit of Rx volume	Weekly brand tracking and decile targeting; small-cell imputation
Decile	Prescribers ranked by (projected) volume into ten groups by volume share	Deciles 8–10 hold most of the volume; cut-offs are vendor conventions
Specialty pharmacy / hub data	Referral → benefits verification → PA → fill status for specialty brands	Your brand only; time-to-fill and abandonment
Formulary / UM intelligence	Plan × drug × tier × restrictions (PA, ST, QL)	The access gate; link to claims via plan IDs (imperfect)
Tokenization	One-way hash of identifiers (name, DOB, sex, ...) so records match without revealing identity	Deterministic vs probabilistic linkage; a licensed service (e.g., Datavant-style)

Part 3 • The data-quality checklist (run before every analysis)

#	Check	How	Typical finding	Action
1	Code inventory	Frequency table of diagnosis and procedure codes	Neighbour codes, symptom codes, unclassified J-codes	Define the code list explicitly; document inclusions
2	Duplicates	Duplicate on (patient, date, provider, code, dx, amount)	1–5% resubmissions	Keep one; log the rule
3	Adjustments	Amount ≤ 0 ; status flags	Reversals, corrections	Exclude from utilization; keep for cost netting
4	Missing keys	Null rate per key column	NPI, POS, plan IDs	Fallbacks (billing NPI); report the rate
5	Referential integrity	Keys not in the master (NPI, NDC, plan)	Retired NDCs, new NPIs	Refresh masters; unmapped must be explained
6	Date sanity	Claims before the vendor dx date; gaps; future dates	Back-dated claims, eligibility backfills	Outlier rules; use claims-derived index, not vendor dates
7	Enrollment / capture	Join eligibility; activity per period	8–15% with gaps or loss	Baseline and follow-up criteria; censor
8	Lag tail	Counts per month; completeness vs. mature months	Last 2–3 months thin	Completeness factors; compare mature periods
9	Small cells	Count patients per reported cell	Cells < 11	Suppress or aggregate before sharing
10	Reconciliation	Totals vs. an independent source (DDD, sales, prior extract)	Panel change, restatement	Explain every gap > a few %

Write a cleaning log

Every exclusion, with counts, in the order applied. It goes in the appendix of every deliverable and is the first thing a reviewer (or a regulator) asks for.

Part 4 • The cohort-definition template (use in your data plan)

Element	Your choice	Trade-off to state
Purpose	Market sizing · targeting · outcomes · access · forecasting	Decides how loose or tight everything below is
Disease / event codes	ICD-10 list (family, not one code); procedure or drug codes if event-defined	Narrow misses; broad admits neighbours
Confirmation rule	≥ 2 claims on different days within N days; or 1 inpatient claim; or a specialist claim	Rule-outs vs. lost patients (sensitivity vs. specificity)
Index date	First qualifying claim; first treatment; first specialist visit	Anchors all time-to-event measures
Baseline (washout)	Continuous enrollment $\geq 180/365$ days before index, gap ≤ 45 days	Incident vs. prevalent; loses the newly insured
Follow-up	Enrollment after index; censor at coverage end / death / extract end	Censoring $\neq$ death
Activity / capture	≥ 1 claim per period	Guards against half-seen patients
Exclusions	Prior disease codes; age; other cancers; clinical trial flags	Say which and why
Attrition table	Counts at every step	Non-negotiable
Validation	Chart review sample; comparison with registry incidence; sensitivity analysis on N and windows	Reports the uncertainty, not just the number

Part 5 • De-identification and what you may do with the data

5.1 HIPAA Privacy Rule — two routes (45 CFR §164.514)

Safe Harbor: remove 18 identifier types — names; geographic subdivisions smaller than a state (except the first three ZIP digits when the ZIP3 population exceeds 20,000); all elements of dates except year (and ages over 89 aggregated to 90+); telephone and fax numbers; email; SSN; medical record numbers; health plan beneficiary numbers; account numbers; certificate/license numbers; vehicle identifiers; device identifiers; URLs; IP addresses; biometric identifiers; full-face photographs; any other unique identifying number or code — and have no actual knowledge the remainder could identify the individual.

Expert Determination: a qualified expert determines and documents that the risk of re-identification is very small, given the anticipated recipient and reasonable methods. Commercial claims and LRx datasets keep exact dates, ZIP3 and prescriber under this route, with controls: tokenized IDs, small-cell suppression, contractual prohibitions on re-identification and on linkage to identified data, and access controls.

5.2 Re-identification risk in plain terms

- Quasi-identifiers combine: ZIP + birth date + sex was enough to uniquely identify ~87% of the U.S. population (Sweeney, 2000). In claims: rare diagnosis × ZIP3 × exact dates × treating NPI.
- Linkage attacks join a de-identified file to a public one (voter rolls, obituaries, news, social media). Each added source shrinks k — the number of people who share a record's profile.
- Model-based inference: models trained on de-identified features can reconstruct inputs or infer sensitive attributes. Defences: aggregation, dimensionality reduction, noise/differential privacy, federated training, monitoring.

5.3 Rules for this course and for the job

Always	Only with governance	Never
Suppress small cells (< 11 is common) in anything shared · Keep the cleaning log and cohort definition with every result · Report capture/coverage next to every rate · Use synthetic or public data for development and demos	Linking two patient-level sources (tokenization vendor + determination) · Prescriber-level reporting (state restrictions, AMA PDRP opt-outs) · Training ML models on licensed data (check the license) · Sharing extracts with partners or agencies	Attempt to re-identify anyone, even to check · Paste patient-level rows into an external AI tool · Use patient-level data to target an individual patient · Contact a prescriber about an identified patient · Keep data after the license ends

Public data you may use freely for the term project: CMS Medicare Part D Prescribers by Provider and Drug; CMS Open Payments; NPPES NPI registry; FDA NDC directory; SEER incidence; CDC WONDER; the course's synthetic datasets.